

Parliamentary Collapse Under Fragmentation

A four-institution agent-based model of legislative passage

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github.com/tofuadmiral/institutional-representation-abm · institutional-representation-abm.streamlit.app

The puzzle

Parliamentary systems pass more bills than presidential systems at baseline, but collapse to near-zero passage when legislatures fragment. Three competing mechanistic accounts exist in the literature: coalition-formation failure, party discipline, and committee gatekeeping.

These three mechanisms operate simultaneously in any real legislature. Observational studies cannot separate their contributions. A simulation can.

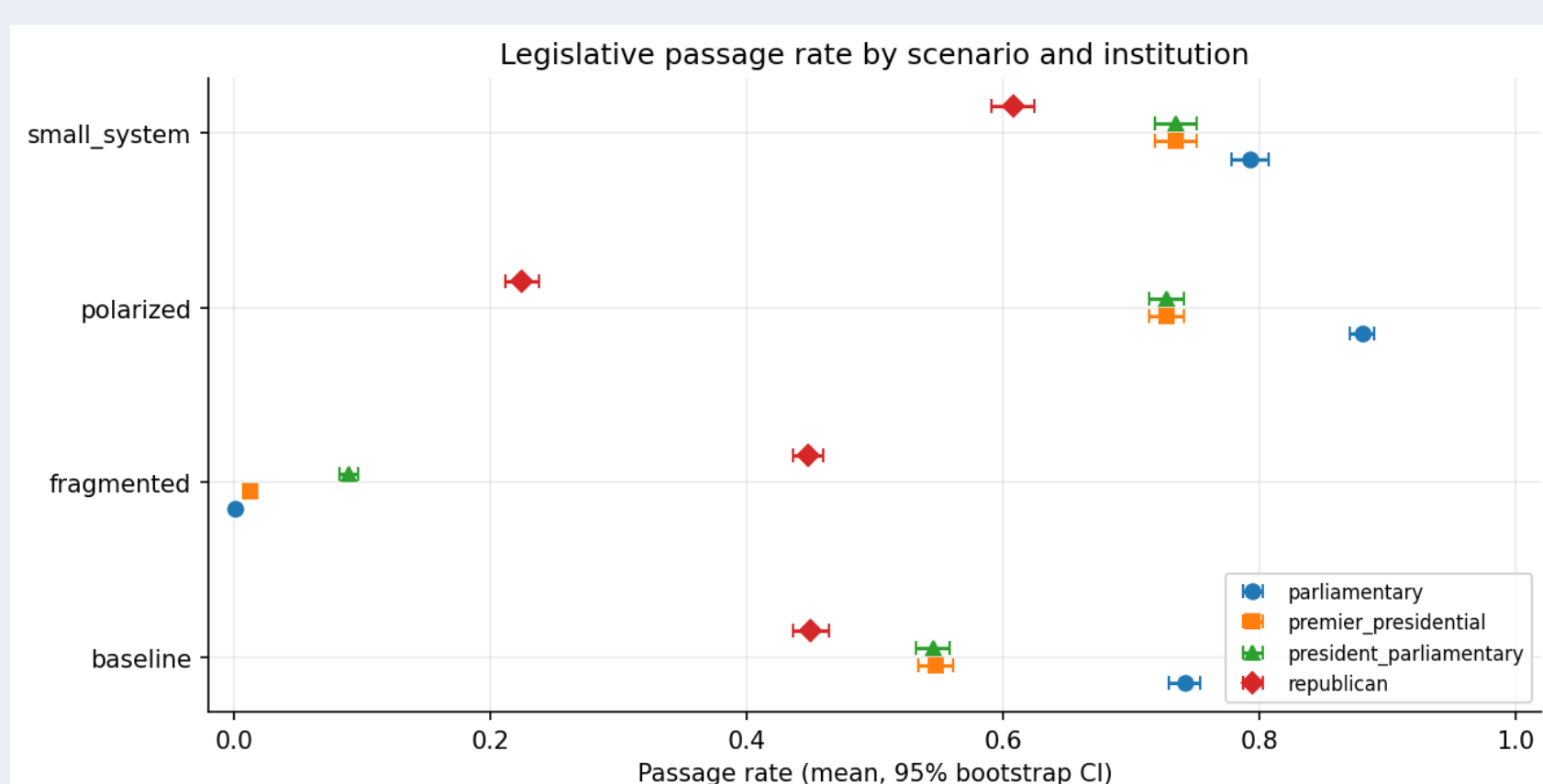
Four institutions, one model

Four democratic legislative systems, shared agent types, different bill-processing pipelines:

- **Parliamentary:** cabinet responsible to parliament, strong discipline, confidence votes, no veto.
- **Republican:** separated executive with veto, 2/3 override, weak discipline.
- **Premier-presidential (France):** majority-driven cabinet, presidential veto, no PM dismissal.
- **President-parliamentary (Russia):** president-driven cabinet, veto, dismissal allowed.

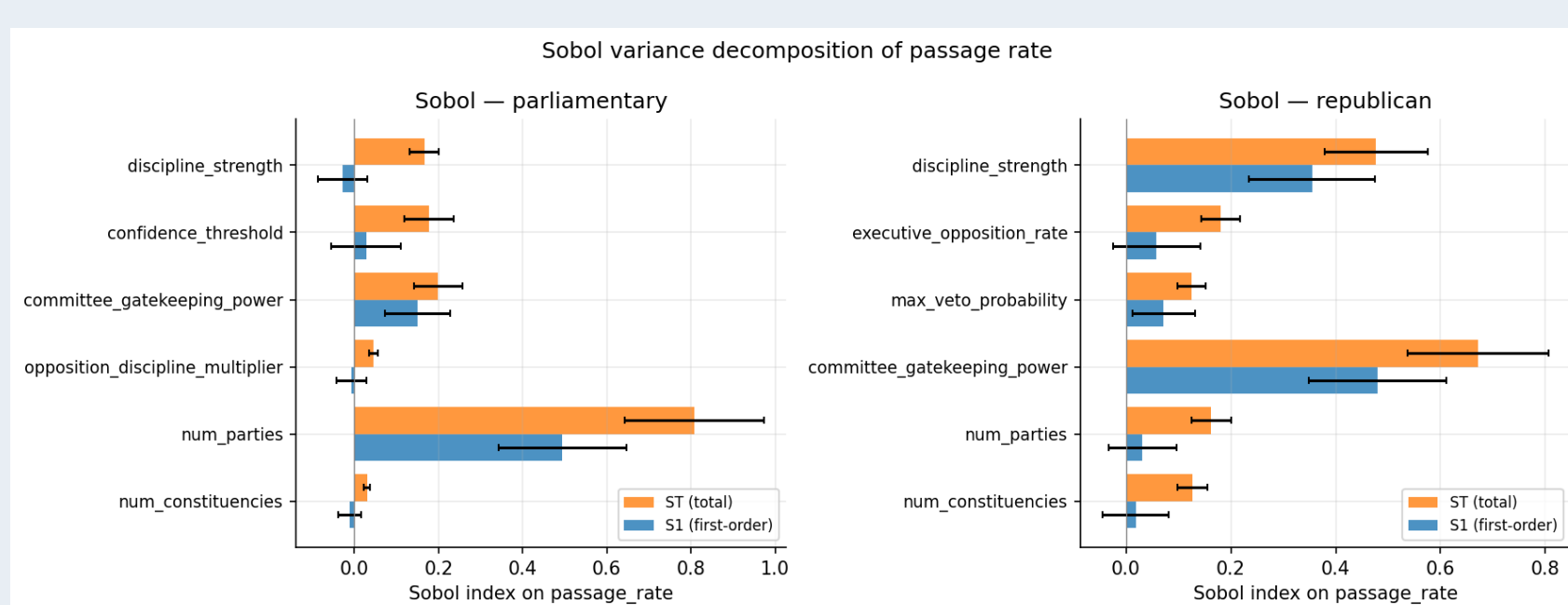
Semi-presidential variants reachable from one class through two configuration toggles. Headline results at $N = 200$ seeds per (scenario, institution) cell with 95% bootstrap CIs (parameter sweeps at $N = 100$).

Headline: passage across scenarios



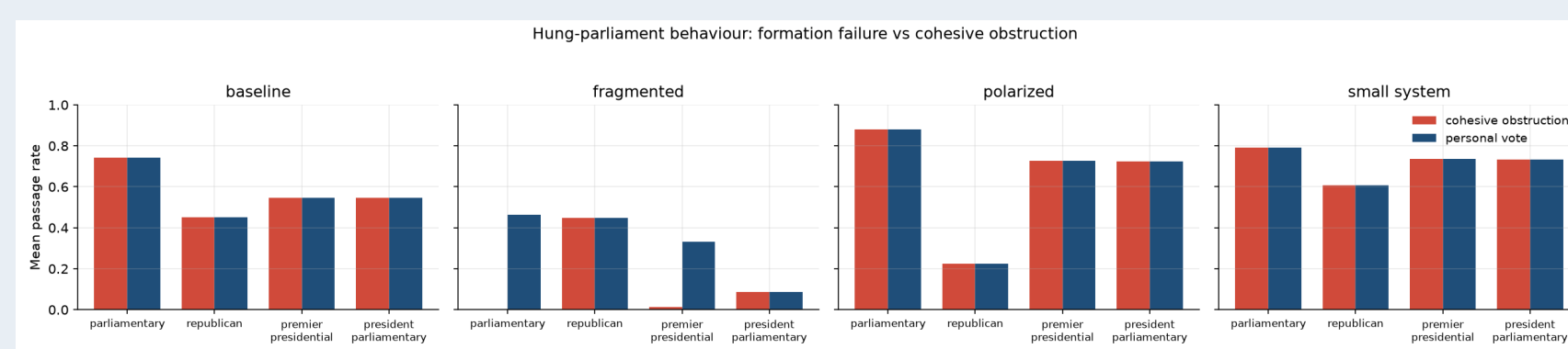
Baseline: parliamentary +29pp over republican (Welch $t = 31.1$, Cohen $d = 3.11$). **Fragmented:** parliamentary and premier-presidential collapse; president-parliamentary partially rescues.

Sensitivity: different bottlenecks



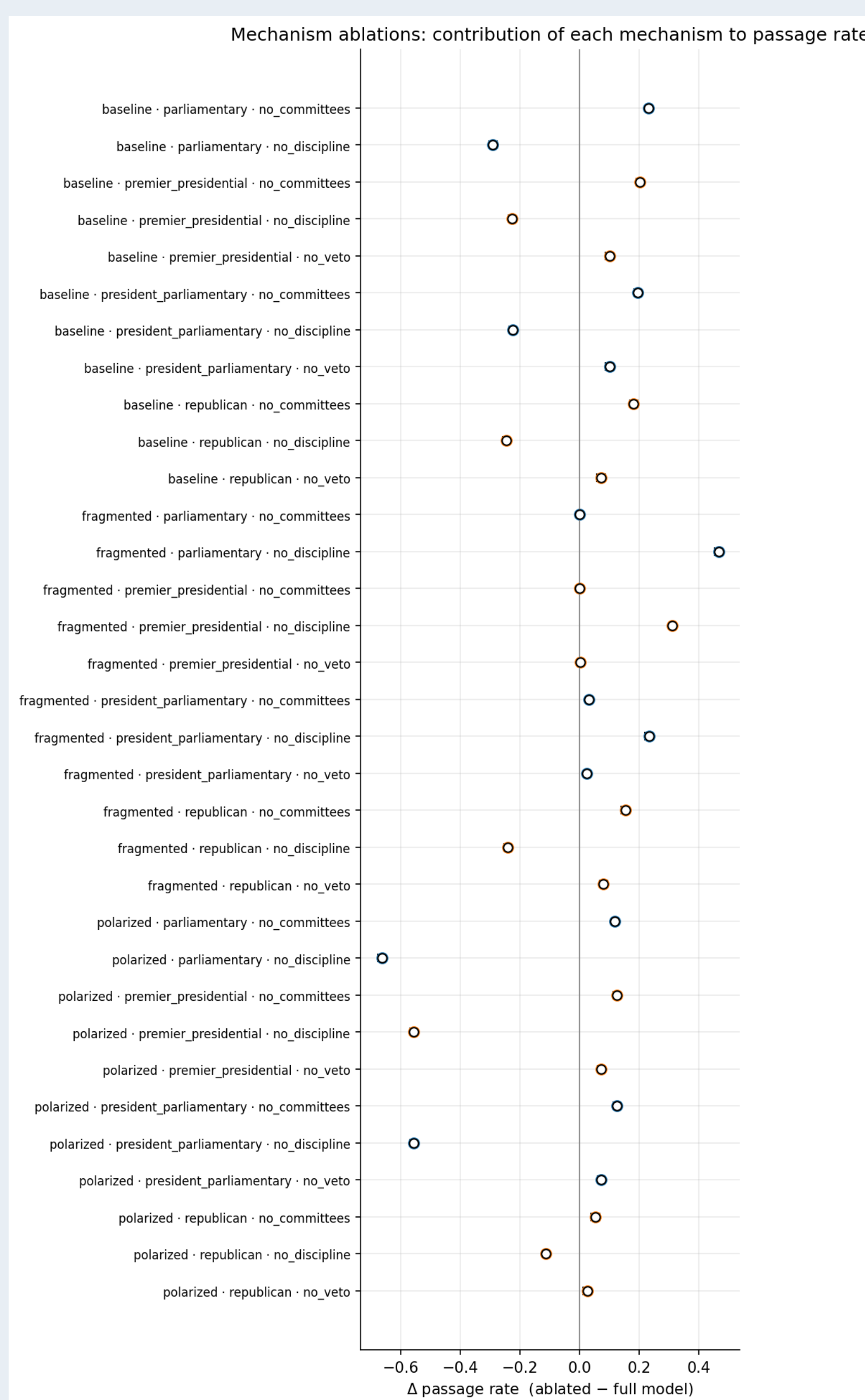
Sobol S_T shows parliamentary passage is dominated by `num_parties` (0.81), republican by `committee_gatekeeping` (0.67). Structurally different bottlenecks.

Finding 1a: hung parliaments still legislate



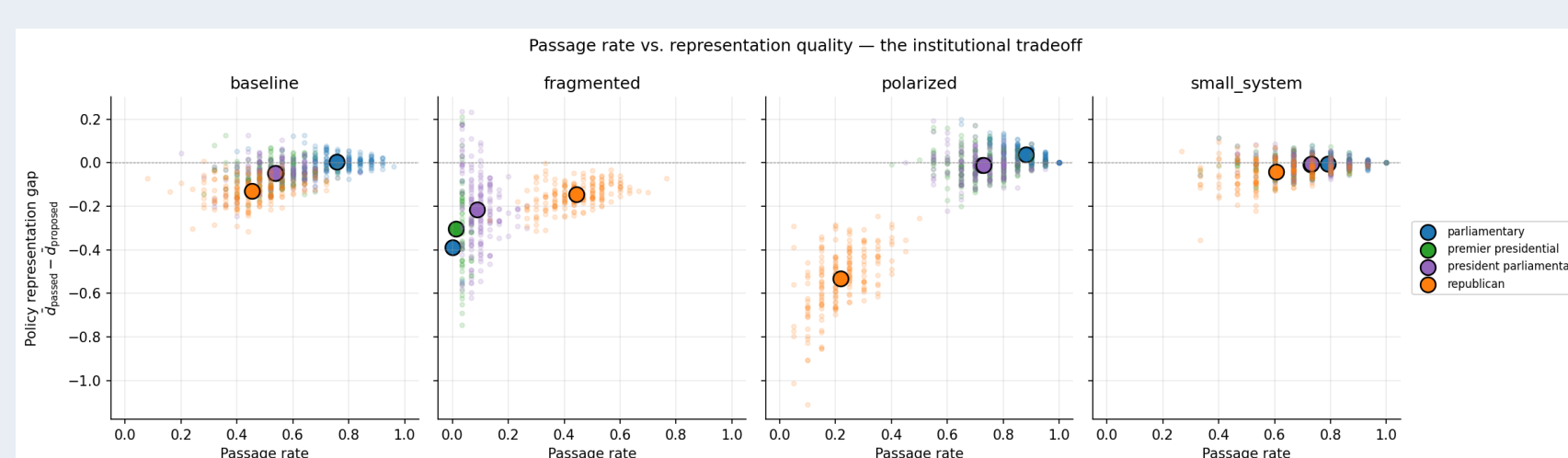
Fragmentation makes government formation impossible, yet passage continues: letting a hung parliament's MPs vote personally keeps parliamentary legislation at 46.4%, indistinguishable from republican's 44.8%. Collapse requires **cohesive opposition obstruction**: every MP whipped against all business drives passage to 0.05%.

Finding 1b: discipline aggregates votes — both ways



Zeroing the whip (“no_discipline”) restores fragmented parliamentary passage from 0.05% to 46.7%; rescue magnitude falls monotonically with each institution's dependence on majority government formation. Under polarisation the same ordering holds with opposite sign: discipline costs parliamentary 66pp when bills are extreme. Discipline aggregates blocs wherever they exist — governing coalitions pass, anti-system oppositions obstruct.

Finding 2: passage–representation tradeoff

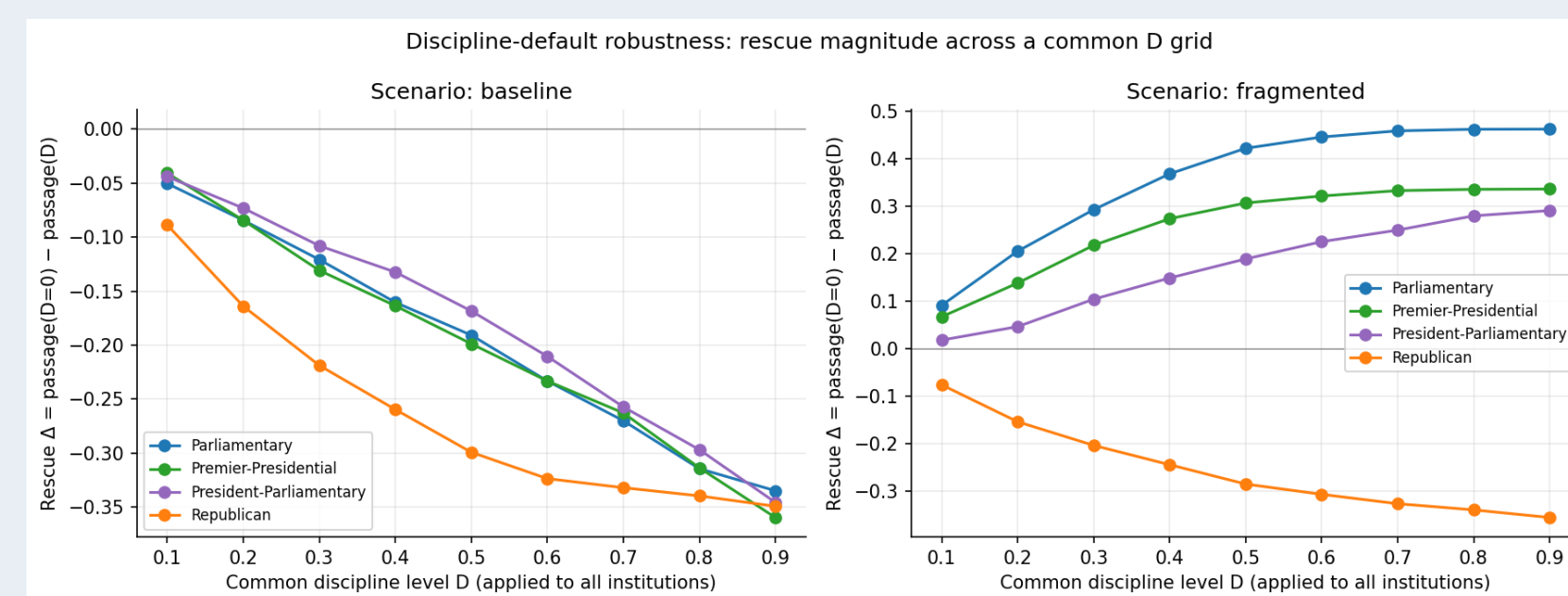


Policy representation gap = $\bar{d}_{\text{passed}} - \bar{d}_{\text{proposed}}$ (L2 distance to constituency median). At polarised:

Institution	Passage	Repr. gap
Parliamentary	0.882	+0.037
Premier-pres.	0.725	−0.012
President-parl.	0.730	−0.010
Republican	0.218	−0.530

Filtering and throughput lie on a single

Robustness: ordering survives parameter sweeps



Set every institution's `discipline_strength` to the same D across $\{0.1, \dots, 0.9\}$. The monotone ordering holds at **9/9 tested values under fragmentation**. The structural claim does not depend on our default parameter choices; a regression test enforces it.

Takeaways

1. **Opposition cohesion drives the collapse.** Fragmented parliaments legislate at presidential rates when MPs vote personally; cohesive anti-system obstruction is what halts them. The pathology lives on the opposition side.
2. **Discipline is the universal vote-aggregation mechanism** for majority-dependent institutions: it blocks passage under obstruction and enables passage under polarisation.
3. **The passage–representation tradeoff is one axis.** Institutions cannot improve throughput and representational fidelity simultaneously; the choice is structural.
4. **Reform implication.** Weakening party discipline is an alternative to electoral reform for fragmented parliaments — but only where oppositions bargain rather than obstruct.

Reproducibility

- Python 3.13, Mesa 2.x, MIT-licensed
- 80 automated tests, GitHub Actions CI
- ODD protocol (Grimm et al. 2020) in repo
- Full paper: `paper/main.pdf`
- Live demo:
institutional-representation-abm.streamlit.app

References

Capoccia (2005). Cox & McCubbins (1993). Laver & Schofield (1990). Linz (1990). Noble & Schulmann (2018). Sartori (1976). Shugart & Carey (1992). Strøm (1984, 1990). Tsebelis (2002).
Full bibliography in `paper/references.bib`.